

CS 767 Advanced Machine Learning Homework 9

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1. **Kernels.** Consider an image of size $(J \times J)$ that is padded with an additional P pixels of value 0 on all sides and which is then convolved with a kernel of size $(2M + 1 \times 2M + 1)$. Argue why choosing $P = M$ will give a feature map the same size as the image.

Because it is not specified, the stride is assumed to be 1. Therefore, the size of the output image is given by the equation $O = I + 2p - h + 1$.

The goal is to make the feature map the same size as the image, $O = I$.

Then the rest of the terms must be zero, $0 = 2p - h + 1$.

Rearrange terms to get $h = 2p + 1$.

Since the kernel size $h = 2M + 1$ and padding size $p = P$, substitute to get $2M + 1 = 2P + 1$.

Simplify to show $M = P$. Therefore, to obtain a feature map the same size as the image, $P = M$ or the padding must be equal to half the kernel size minus one, $p = \frac{h-1}{2}$.

2. **More kernels.** Explicitly calculate the output of the following convolution of a (4×4) input image with a (2×2) filter.

2	5	-3	0
0	6	0	-4
-1	-3	0	2
5	0	0	3

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-2	0
4	6

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32	14	-18
-22	-24	12
22	6	18

$$\begin{aligned}
 (2 \times -2) + (5 \times 0) + (0 \times 4) + (6 \times 6) &= -4 + 0 + 0 + 36 = 32 \\
 (5 \times -2) + (-3 \times 0) + (6 \times 4) + (0 \times 6) &= -10 + 0 + 24 + 0 = 14 \\
 (-3 \times -2) + (0 \times 0) + (0 \times 4) + (-4 \times 6) &= 6 + 0 + 0 - 24 = -18 \\
 (0 \times -2) + (6 \times 0) + (-1 \times 4) + (-3 \times 6) &= 0 + 0 - 4 - 18 = -22 \\
 (6 \times -2) + (0 \times 0) + (-3 \times 4) + (0 \times 6) &= -12 + 0 - 12 + 0 = -24 \\
 (0 \times -2) + (-4 \times 0) + (0 \times 4) + (2 \times 6) &= 0 + 0 + 0 + 12 = 12 \\
 (-1 \times -2) + (-3 \times 0) + (5 \times 4) + (0 \times 6) &= 2 + 0 + 20 + 0 = 22 \\
 (-3 \times -2) + (0 \times 0) + (0 \times 4) + (0 \times 6) &= 6 + 0 + 0 + 0 = 6 \\
 (0 \times -2) + (2 \times 0) + (0 \times 4) + (3 \times 6) &= 0 + 0 + 0 + 18 = 18
 \end{aligned}$$

3. **The CIFAR-10 Dataset.** Repeat the CIFAR-10 dataset analysis from the previous homework. However, this time, train a convolutional network on the dataset. You will be using Categorical Cross Entropy (see Sec. 5.4.4 and sec. 6.4.3 in Chris Bishop's book) as your loss function, along with a SoftMax activation layer as the final layer of your CNN. Therefore, use the Keras utility function `to_categorical()` to one-hot encode your labels. Further, normalize all pixel values within the range $[0,1]$ as a preprocessing step.

Study the Keras 2D convolutional layer thoroughly before constructing your network. The network has the following specifications.

- a. An input layer of size $32 \times 32 \times 3$ (since the images have 3 channels, viz, red, blue, and green).
- b. First convolutional + maxpooling block: 32 filters of size 3×3 each. Use “same” padding to preserve output size for *all* convolutional layers in this network and ReLU activations. Use maxpooling blocks of size (2×2) throughout your network. Let's stick with stride = 1 everywhere for simplicity.
- c. Second convolutional + maxpooling block: 64 filters of size 3×3 each.
- d. Third convolutional + maxpooling block: 128 filters of size 3×3 each.
- e. flatten the output of this layer and feed it to a dense layer with 128 units (and ReLU activation).
- f. The final layer is a fully connected layer with SoftMax activation and 10 outputs (one for each class of CIFAR10).

Compile the model with the Adam optimizer and show the Accuracy vs Epoch plot as before. How many parameters does this model have? How does it compare with your earlier feedforward network? What are the final train and test accuracy values this model achieves? How do they compare with your earlier feedforward network?